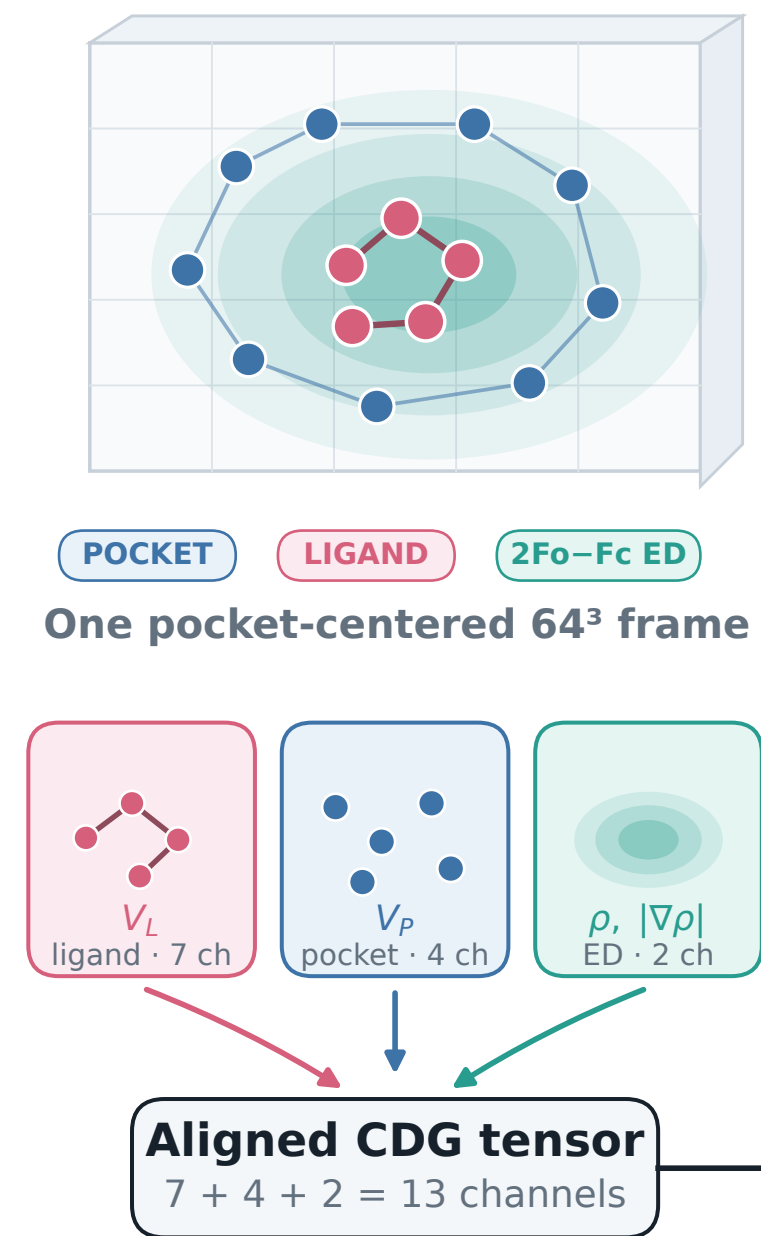


Single-encoder CDG pipeline

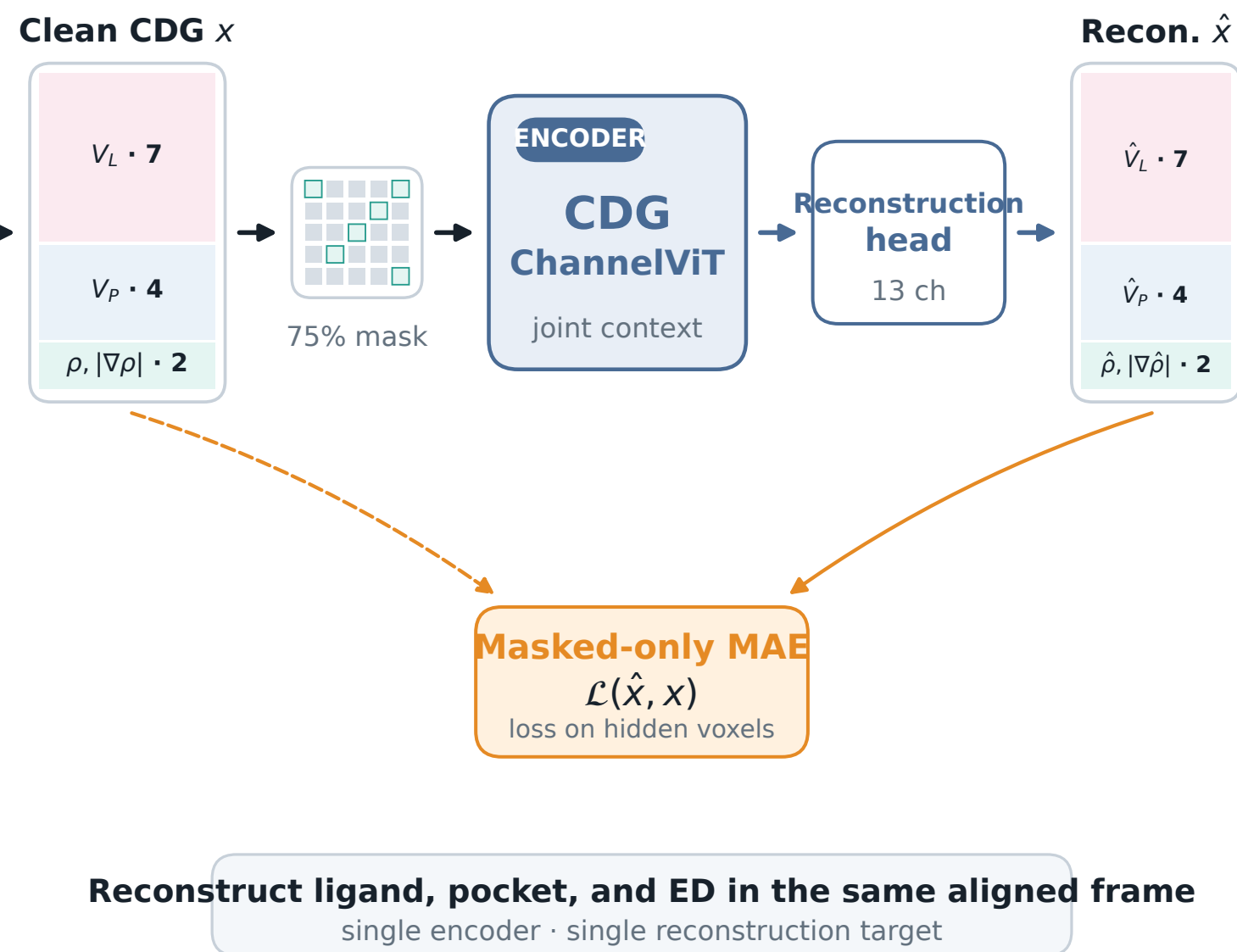
Pocket coordinates, ligand coordinates, and experimental electron density are learned jointly in one spatial encoder

(a) Aligned input data



C: coordinates · D: density · G: density gradient

(b) Joint CDG pre-training



(c) High-level VoxBind fusion

